

SUDARSHAN UAV — OPERATOR TRAINING COURSE

Five-Module Structured Curriculum | Issue 1.0

This course prepares operators to safely configure, operate, and maintain the SUDARSHAN quadrotor UAV system. Completion of all five modules and a score $\geq 23/25$ on the Operator Certification Quiz is required for independent flight operations.

MODULE 1: System Orientation

Learning Objectives

- Identify all major hardware components and their roles in the SUDARSHAN system.
- Trace the power flow from LiPo battery through BEC, ESCs, and microcontrollers.
- Describe the three-layer software architecture (FC / Uno driver / ESP32 gateway).

Content Overview

The SUDARSHAN quadrotor uses an Arduino Mega as the primary Flight Controller (FC), an Arduino Uno as the motor driver board, and an ESP32 as the wireless gateway. A 3S LiPo (11.1 V nominal) powers four brushless motors via individual ESCs. Power is distributed through a PDB; a 5 V BEC supplies the microcontrollers. An MPU-6050 IMU provides roll/pitch/yaw data via I2C. An HC-SR04 ultrasonic sensor measures altitude up to ~2 m. The PCA9685 PWM driver on the Uno board controls ESC signal lines on channels CH0–CH3.

Layer Architecture: Layer 1 (FC / Mega) runs the PID control loop at 250 Hz. Layer 2 (Uno) receives 10-byte motor packets and outputs calibrated PWM to ESCs. Layer 3 (ESP32) bridges TCP Wi-Fi commands from the GCS to the Mega via UART with a 5 V $\rightarrow$ 3.3 V voltage divider on the Mega TX line.

Exercise 1 — Wiring Table Completion

Fill in the blank cells in the table below using the SUDARSHAN wiring diagram.

Signal / Component	Connected To	Function	Notes
MPU-6050 SDA	Mega A4	I2C Data	
MPU-6050 SCL	Mega A5	I2C Clock	
PCA9685 SDA	Uno A4	I2C Data	
PCA9685 SCL	Uno A5	I2C Clock	
Mega TX1	_____	Motor commands	UART 115200
Uno RX	Mega TX1	_____	UART 115200
ESP32 RX	_____	GCS commands	Voltage divider
Mega TX2	_____	ESP32 bridge	
HC-SR04 TRIG	Mega D9	_____	
HC-SR04 ECHO	Mega D10	Ultrasonic echo	

Module 1 Quiz

Q1.	The MPU-6050 communicates with the Mega via which protocol and pins?
Q2.	Name the three software layers of the SUDARSHAN architecture and the microcontroller responsible for each.
Q3.	What is the purpose of the 5 V $\rightarrow$ 3.3 V voltage divider on the Mega TX2 line?

MODULE 2: GCS Operation

Learning Objectives

- Connect the GCS laptop to the SUDARSHAN Wi-Fi access point.
- Navigate all five GCS tabs and describe the purpose of each.
- Interpret all telemetry fields and the status bar indicators.

Content Overview

The Python GCS application connects to the ESP32 at TCP 192.168.4.1:5760. The interface is organised into five functional tabs:

Tab	Purpose
FLIGHT	ARM / DISARM, HOVER, LAND, KILL commands; live telemetry panel
PRESET	Load, edit, and transmit PRESET mission segments (bearing + distance)
PREFLIGHT	Run MOTOR_TEST, IMU preflight check, sonar sanity test
NAV	IMUMISSION start / abort, override waypoint injection
GUIDE	Manual OVERRIDE mode with on-screen virtual joystick

Key telemetry fields: roll_deg, pitch_deg, yaw_deg (IMU angles), alt_cm (sonar altitude), bat_mv (battery mV), sonar_ok (1=valid), state (IDLE/ARMED/HOVER/LAND/KILL), dms_countdown (seconds until DMS timeout).

Exercise 2 — Simulated GCS Session Checklist

Work through the following 10-step procedure and tick each step as completed.

■ Step 1.	Connect laptop Wi-Fi to SSID 'SUDARSHAN_AP'
■ Step 2.	Launch GCS application — verify connection indicator turns GREEN
■ Step 3.	Open PREFLIGHT tab — run MOTOR_TEST with props removed
■ Step 4.	Confirm all 4 motor channels respond in sequence (0→25% duty)
■ Step 5.	Open PREFLIGHT tab — run IMU_CHECK; confirm pass (green)
■ Step 6.	Open PREFLIGHT tab — run SONAR_CHECK; confirm alt_cm > 0
■ Step 7.	Check bat_mv > 10500 mV in telemetry panel
■ Step 8.	Open FLIGHT tab — issue ARM command; confirm state = ARMED
■ Step 9.	Issue HOVER; observe alt_cm stabilises within ±5 cm
■ Step 10.	Issue LAND; confirm state transitions LAND → DISARMED

Module 2 Quiz

Q1.	Which TCP port and IP address does the GCS use to communicate with the ESP32?
Q2.	Describe the difference between the PRESET tab and the NAV tab.
Q3.	What telemetry field indicates whether the sonar sensor is providing valid data?

MODULE 3: Flight Operations

Learning Objectives

- Execute the full pre-flight checklist without reference material.
- Perform ARM, HOVER, controlled LAND, and emergency procedures correctly.
- Select the appropriate emergency command for each abnormal condition.

Pre-Flight Checklist

MECHANICAL

- All 4 props tight and correct orientation (CW/CCW)
- Frame arms secure, no cracks
- Landing gear intact
- Motor shafts free — no binding

ELECTRICAL

- LiPo charged ($> 11.8\text{ V} = 100\%$)
- Balance connector checked
- All connectors fully seated
- No damaged wiring

FIRMWARE

- Mega FC serial shows READY
- Uno serial shows READY
- ESP32 AP visible in Wi-Fi list

GCS

- GCS connected (green indicator)
- MOTOR_TEST passed (no props)
- IMU preflight passed
- Sonar sanity passed
- bat_mv $> 10\,500\text{ mV}$ confirmed

AREA

- 10 m safety perimeter clear of persons
- Wind $< 5\text{ m/s}$
- No overhead obstructions within flight path

Emergency Decision Table

Exercise 3: Given each telemetry state below, choose and record the correct command.

Scenario	Your Command	Reference Answer
Uncommanded CW yaw rotation, state = HOVER	_____	KILL / check props
bat_mv = 9800, state = HOVER	_____	Auto-LAND triggered

Scenario	Your Command	Reference Answer
sonar_ok = 0, state = HOVER	_____	LAND manually
No GCS telemetry for 10 s, state = ARMED	_____	DMS triggers LAND
Drone drifts rapidly toward obstacle, HOVER	_____	KILL then power-cycle

Module 3 Quiz

Q1.	At what battery voltage does the FC automatically issue a LAND command?
Q2.	If the drone begins drifting toward an obstacle during HOVER and GCS is unresponsive, what is the last-resort action?
Q3.	State the correct power-on sequence for the SUDARSHAN system.

MODULE 4: Mission Planning

Learning Objectives

- Construct a valid PRESET mission using bearing and distance segments.
- Explain the three IMUMISSION execution phases: SEG_TURN, SEG_FLY, SEG_PAUSE.
- Identify conditions that trigger mission abort and execute the abort procedure.

PRESET Segment Format

Each PRESET segment is defined by two parameters: **bearing** (0–359°, relative to current IMU heading at mission start) and **distance** (metres, 0.5–20 m). Segments are executed in order. The drone uses the complementary-filter yaw to track heading and motor differential thrust for forward movement.

IMUMISSION Execution Phases

Phase	Description	Completion Criterion	Typical Duration
SEG_TURN	Drone rotates to the target bearing	Yaw PID drives heading error < 3°	Typically 1–3 s
SEG_FLY	Drone translates forward at 0.4 m/s	Distance estimated from integration	Variable
SEG_PAUSE	Drone hovers in place for 2 s	Altitude hold active	2 s fixed

Abort Conditions

- sonar_ok = 0 during any flight phase → immediate LAND
- bat_mv ≤ 9900 mV during mission → LAND and mission abort
- Operator issues LAND or KILL command → mission halted, drone responds
- DMS timeout (30 s no GCS heartbeat) → LAND

Exercise 4 — Design a 3-Segment Square Mission

Complete the PRESET table below to fly a 5 m square (north start, clockwise). Fill in the bearing for each segment.

Seg #	Bearing	Distance	Execution Phases
1	0° (North)	5 m	SEG_TURN → SEG_FLY → SEG_PAUSE
2	_____	5 m	SEG_TURN → SEG_FLY → SEG_PAUSE (answer: 90°)
3	_____	5 m	SEG_TURN → SEG_FLY → SEG_PAUSE (answer: 180°)

Module 4 Quiz

Q1.	In SEG_FLY phase, how does the FC estimate distance travelled?
Q2.	What parameter must be set correctly at mission start for bearing references to be valid?
Q3.	List three conditions that cause an automatic IMUMISSION abort.

MODULE 5: Maintenance & Troubleshooting

Learning Objectives

- Perform ESC calibration and motor-ID wizard correctly.
- Adjust PID gains safely using the GCS tuning interface.
- Diagnose common faults from symptom descriptions.

ESC Calibration Procedure

1. Remove all propellers.
2. Power on Mega + Uno. In GCS, open PREFLIGHT → ESC_CALIB.
3. GCS sends max-throttle signal to all ESCs simultaneously.
4. Connect LiPo — ESCs enter calibration mode (single beep).
5. GCS sends min-throttle — ESCs confirm calibration (descending beeps).
6. Power-cycle ESCs; verify linear throttle response with MOTOR_TEST.

Motor-ID Wizard

The wizard spins each motor individually at 15% throttle for 0.5 s. Observe the spinning motor and verify it matches the layout (FL=CH0, FR=CH1, RL=CH2, RR=CH3). If mismatched, update MOTOR_MAP in firmware or re-run SET_MOTOR_MAP command from GCS.

PID Tuning Guidelines

Symptom	Diagnosis	Action
Oscillation at high frequency	Kp too high	Reduce Kp by 10–15%
Slow response, large overshoot	Kp too low	Increase Kp by 10%
Steady-state error remains	Ki too low	Increase Ki by 5%
Integrator windup / drift	Ki too high	Reduce Ki; check for noise
Noise / jitter on actuators	Kd too high	Reduce Kd or add D-filter

Firmware Update Procedure

1. Disconnect Uno Pin 0/1 from Mega before flashing Uno.
2. Use Arduino IDE with correct board/port selected.
3. Flash Mega FC firmware first; verify READY on serial.
4. Flash Uno motor driver firmware; verify READY.
5. Flash ESP32 using PlatformIO; verify AP is visible.
6. Run full PREFLIGHT test after each firmware update.

Exercise 5 — Fault Identification

Match each symptom to its most likely fault. Write the fault number in the answer column.

Symptom	Answer	Fault Library
Drone spins clockwise with no yaw input	___	1. Reversed motor direction / wrong prop
IMU shows 0° roll/pitch even when tilted	___	2. I2C address conflict / wiring fault

Symptom	Answer	Fault Library
Motors spin at unequal speeds during MOTOR_TEST	—	3. ESC not calibrated / motor map wrong
GCS shows bat_mv = 0 constantly	—	4. Voltage divider missing / ADC pin float
LAND mode never detects ground (<8 cm trigger)	—	5. Sonar pointing at angle / blocked

Module 5 Quiz

Q1.	Why must Uno Pins 0 and 1 be disconnected before flashing firmware?
Q2.	A pilot observes high-frequency oscillation in roll during HOVER. Which PID parameter should be reduced first?
Q3.	What is the purpose of the motor-ID wizard and when should it be run?

CERTIFICATE OF COMPLETION
SUDARSHAN UAV Operator Training Course

This certifies that

has successfully completed all five modules of the
SUDARSHAN UAV Operator Training Course
and demonstrated competency in all assessed areas.

Date Completed: _____

Authorized by:

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